

Automatic Tuning and Calibration in ESI Mode

This chapter describes how to tune and calibrate the LTQ Series MS. First, tune and calibrate the instrument in ESI mode—that is, set up the instrument for ESI mode (see [“Installing the ESI or HESI-II Probe”](#) on [page 40](#)). Then, set up the scan parameters for a full-scan, single-stage mass analysis. For most applications, you infuse a calibration solution directly into the API source while running the automatic tune and automatic calibration options provided within the data system.

Generally, you must calibrate the instrument every 1 to 3 months of operation for optimum performance over the entire mass range of the detector.

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Plumbing Connections

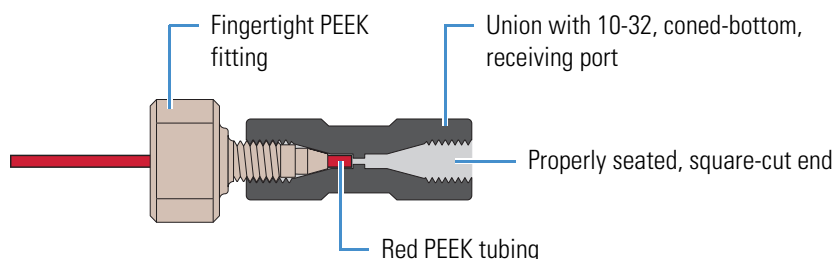
The divert/inject valve is a six-port, two-position, Rheodyne™ injection valve. The ports use standard 10-32 fittings for high-pressure and 1/16 in. OD tubing. To connect the high-pressure tubing to the valve, use the one-piece fingertight fittings provided in the MS Accessory Kit (P/N 97055-62055).

IMPORTANT To help ensure spray stability, make sure that all PEEK tubing is not crimped, kinked, or otherwise damaged.

Ensure the following when you make the plumbing connections:

- The ends of the PEEK tubing are squarely cut (Figure 37). For best results, use a polymeric tubing cutter to ensure square cuts. Poorly cut tubing can cause flow restrictions.
- The PEEK tubing makes contact with the bottom of the receiving port (Figure 37). Tubing that is not properly seated can add dead volume to a chromatographic system.
- The fittings are not overtightened. Tighten the PEEK fittings by using your fingers only, not a wrench. Overtightening the PEEK fittings can cause leaks.

Figure 37. Proper connection for the PEEK tubing and fitting (syringe adapter assembly)



Setting Up the Syringe Pump

Use the syringe pump to directly infuse solution for tuning and calibration.

IMPORTANT To minimize the possibility of cross-contamination, do the following:

- Use a different syringe and length of PEEK tubing for each type of solution.
- Wipe off the needle tip with a clean, lint-free tissue before reinserting the syringe into the syringe adapter assembly.

❖ To set up the syringe pump for direct infusion



CAUTION Sharp object. The syringe needle can puncture your skin. Handle it with care.

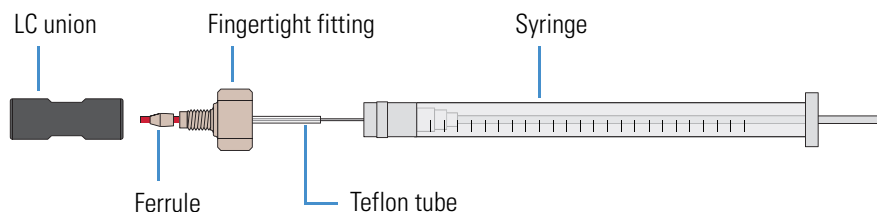
1. Load a clean, 500 μL Unimetrics™ syringe with 450 μL of the ESI calibration solution.
The Chemical Kit contains the ESI calibration solution.

IMPORTANT To minimize the possibility of cross-contamination of the assembly, be sure to wipe off the needle tip with a clean, lint-free tissue before reinserting it into the syringe adapter assembly.

2. Hold the plunger of the syringe in place and carefully insert the tip of the syringe needle into the end of a 4 cm (1.5 in.) Teflon tube with a fingertight fitting and a ferrule to the (black) LC union (Figure 38).

The LC union has a 10-32, coned-bottom receiving port.

Figure 38. Plumbing connection for the syringe



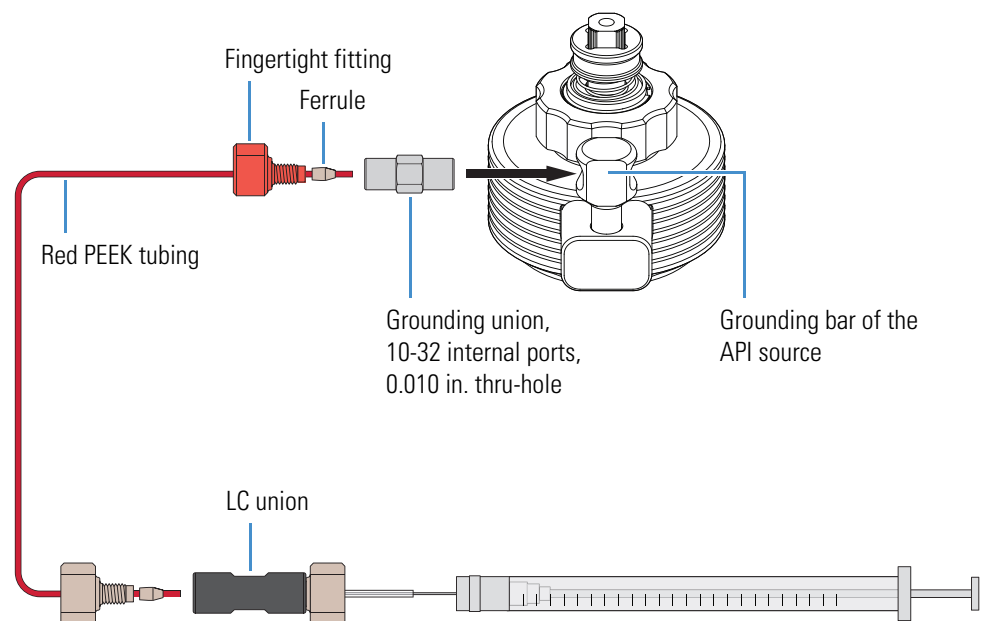
3. Place the syringe into the syringe holder of the syringe pump.
4. Use a fingertight fitting and ferrule to connect an infusion line of red PEEK tubing (0.005 in. ID, 1/16 in. OD) between the LC union and the grounding union held by the grounding bar of the API source.

Figure 39 shows the suggested fittings and ferrules for connecting the LC union to the grounding union. Both the LC union and the grounding union have 10-32, coned-bottom receiving ports.

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Setting Up the Mass Spectrometer for Tuning and Calibration

Figure 39. Plumbing connection between the LC union and the grounding union



5. While squeezing the blue release button on the syringe pump handle, push the handle forward until it just contacts the syringe plunger.

Setting Up the Mass Spectrometer for Tuning and Calibration

Before running the automatic calibration procedure on the LTQ Series MS, manually tune with calibration solution to establish a stable spray of solution and to ensure that the mass analyzer is transmitting a sufficient level of ions to the mass detector. After establishing these conditions, calibrate the instrument automatically to optimize the parameters that affect ion detection.



CAUTION Before beginning normal operation of the mass spectrometer each day, verify that there is sufficient nitrogen for the ESI source. If you run out of nitrogen, the mass spectrometer automatically turns off to prevent atmospheric oxygen from damaging the source. The presence of oxygen in the source when the mass spectrometer is on can be unsafe. In addition, if the mass spectrometer turns off during an analytical run, you might lose data.

Follow these procedures:

1. [To open a tune method for ESI mode](#)
2. [To view the pre-tune ESI source settings](#)
3. [To define the scan parameters](#)
4. [To set the data type and ion polarity mode](#)

❖ **To open a tune method for ESI mode**

1. Open the Tune Plus window (see [page 32](#)).
2. Click the **Open** button.
3. Browse to the **C:\Thermo\Instruments\LTD\methods** folder, and then select either a recently saved tune file that worked on the instrument (preferred), or use the default tune file, Default_ESI.
4. Click **Open**.



The tune method parameters download to the instrument.

❖ **To view the pre-tune ESI source settings**

1. In the Tune Plus window, choose **Setup > ESI Source**.
2. Observe the settings.
3. Click **OK** when finished.

❖ **To define the scan parameters**



1. In the Tune Plus window, click the **Define Scan** button to open the Define Scan dialog box ([Figure 40](#)).

Figure 40. Define Scan dialog box

n	Parent Mass (m/z)	Act. Type	Iso. Width (m/z)	Normalized Collision Energy	Act. Q	Act. Time (ms)
2		CID	1.0	0.0	0.250	30.00

#	Center Mass (m/z)	Width (m/z)
1	609.20	2.00
2		

2. Under Scan Description, do the following:

- In the Mass Range list, select **Normal**.
- In the Scan Rate list, select **Normal**.
- In the Scan Type list, select **Full**.

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Setting Up the Mass Spectrometer for Tuning and Calibration

3. Under Scan Time, do the following:
 - In the Microscans box, enter **1**.
 - In the Max. Inject Time (ms) box, enter **10.000**.
4. Under Source Fragmentation, do one of the following:

Note You must select the Source Fragmentation check box to produce the calibration peak at m/z 138, which is a fragment of caffeine m/z 195.

- For the Velos Pro MS, select the **On** check box.

Note Selecting the Source Fragmentation check box allows collision-induced fragmentation to occur in the API source. Because collision-induced fragmentation performed in the API source is not very specific, this feature is rarely used.

5. Under Scan Ranges, do the following:
 - a. In the Input list, select **From/To**.
 - b. In the First Mass (m/z) column, type **150.00**.
 - c. In the Last Mass (m/z) column, type **2000.00**.
6. Make sure the settings in the Define Scan dialog box are the same as those shown in [Figure 40](#) on [page 67](#).
7. Click **OK**.

❖ To set the data type and ion polarity mode

Centroid 

Profile 

Positive polarity 

Negative polarity 

1. In the Tune Plus window, click the **Centroid/Profile** button to select the **profile** data type.
2. Click the **Positive/Negative** button to select the **positive** ion polarity mode.

This completes the setup to tune with the calibration solution in ESI mode.

Testing the Mass Spectrometer in ESI Mode

To test the operation of the LTQ Series MS, infuse the calibration solution into the ESI source and monitor the real-time display of the mass spectrum of the calibration solution.

IMPORTANT Do not infuse the calibration solution at syringe pump flow rates above 10 $\mu\text{L}/\text{min}$. If you use high concentrations of the Ultramark 1621 component, you can contaminate the sample.

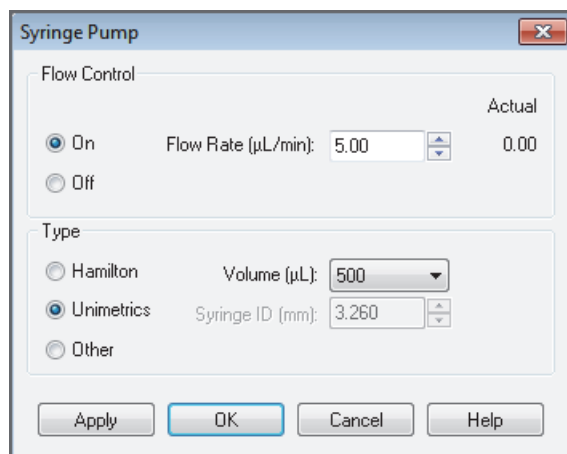
Follow these procedures:

1. To infuse the calibration solution into the API source
2. To monitor the mass spectrum and the spray current

❖ **To infuse the calibration solution into the API source**

1. In the Tune Plus window, choose **Setup > Syringe Pump** to open the Syringe Pump dialog box (Figure 41).

Figure 41. Syringe Pump dialog box



2. Under Flow Control, do the following:
 - a. Select the **On** option.
 - b. In the Flow Rate ($\mu\text{L}/\text{min}$) box, enter **5.00**.

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Testing the Mass Spectrometer in ESI Mode

3. Under Type, do the following:

a. Select the type of syringe.

A 500 μL Unimetrics syringe is supplied with the LTQ Series MS. If you select the Hamilton™ or Unimetrics option, the Syringe ID (mm) box remains unavailable; the data system uses a predefined syringe ID for these syringes. If you select the Other option, the Syringe ID (mm) box becomes available and the Volume (μL) list becomes unavailable.

b. Do one of the following:

- For a Hamilton or Unimetrics syringe, in the Volume (μL) list, select the volume of the syringe.
- For another type of syringe, in the Syringe ID (mm) box, enter the ID of the syringe.

4. Click **OK**.

The syringe pump starts. Or, use the On/Off syringe pump button.

Before starting the next procedure, make sure that you have set up the syringe pump to infuse the ESI calibration solution as described in [“To infuse the calibration solution into the API source”](#) on [page 69](#).

❖ To monitor the mass spectrum and the spray current



1. In the Tune Plus window, click the **On/Standby** button to select the **On** mode.

After you turn on the instrument, it begins scanning, nitrogen flows into the ESI API source, high voltage is applied to the ESI API source, and a real-time mass spectrum display appears in the Spectrum view of the Tune window. For information about spectrums, refer to the topic “Spectrum View” in the data system Help.



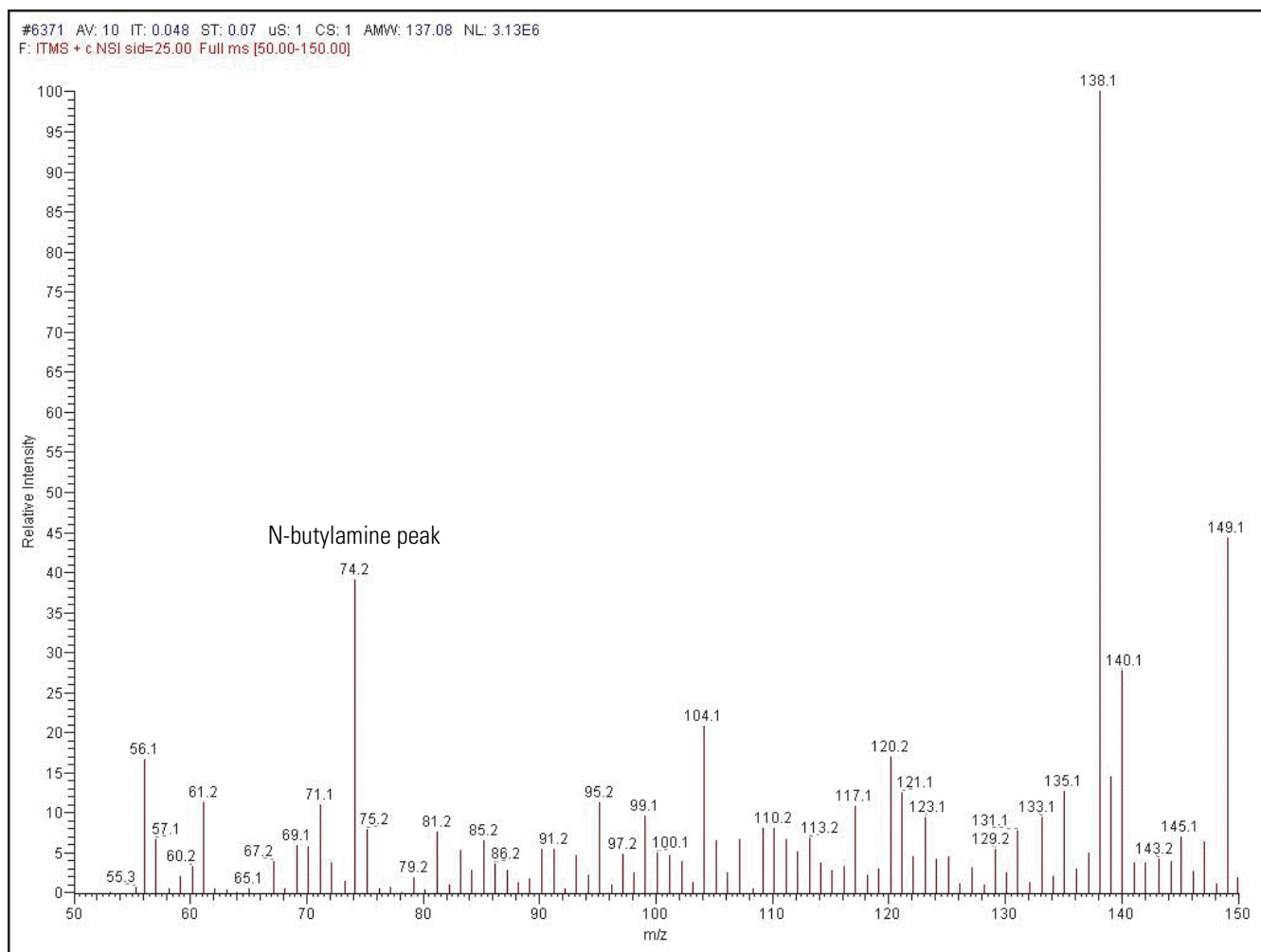
2. Click the **Display Spectrum View** button.

3. Observe the mass spectra of the singly-charged ions in the calibration solution ([Figure 42](#), [Figure 43](#), and [Figure 44](#) on [pages 71 to 73](#)).

The ions are as follows:

- Caffeine: m/z 138.1 and 195 for the Velos Pro MS; m/z 195 for the LTQ XL and LXQ MSs
- MRFA (met-arg-phe-ala acetate salt): m/z 524
- Ultramark 1621: m/z 1022, 1122, 1222, 1322, 1422, 1522, 1622, 1722, 1822, and 1922

Figure 42. Spectrum of the calibration solution for the Velos Pro MS (lower range)



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Figure 43. Spectrum of the calibration solution for the Velos Pro MS (upper range)

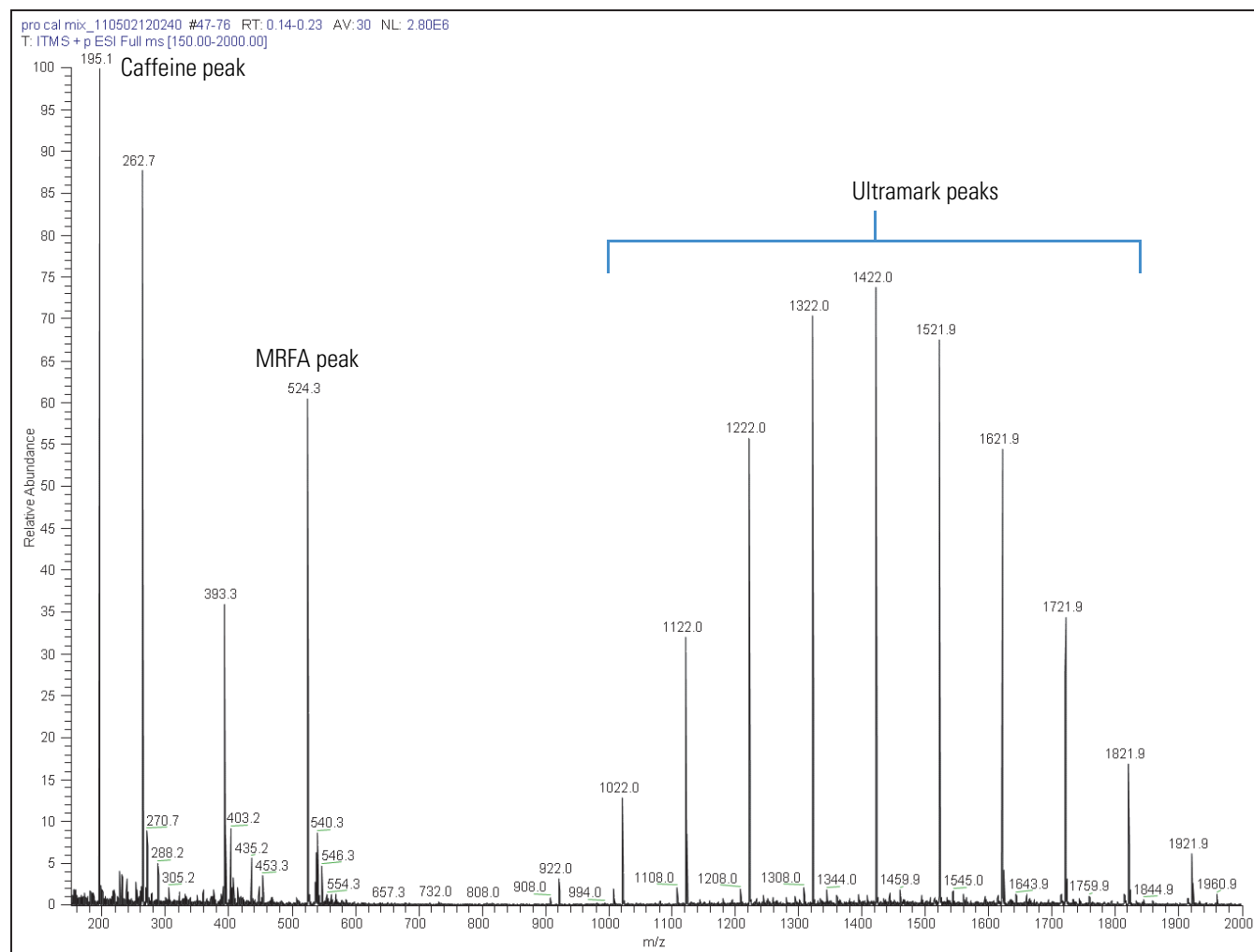
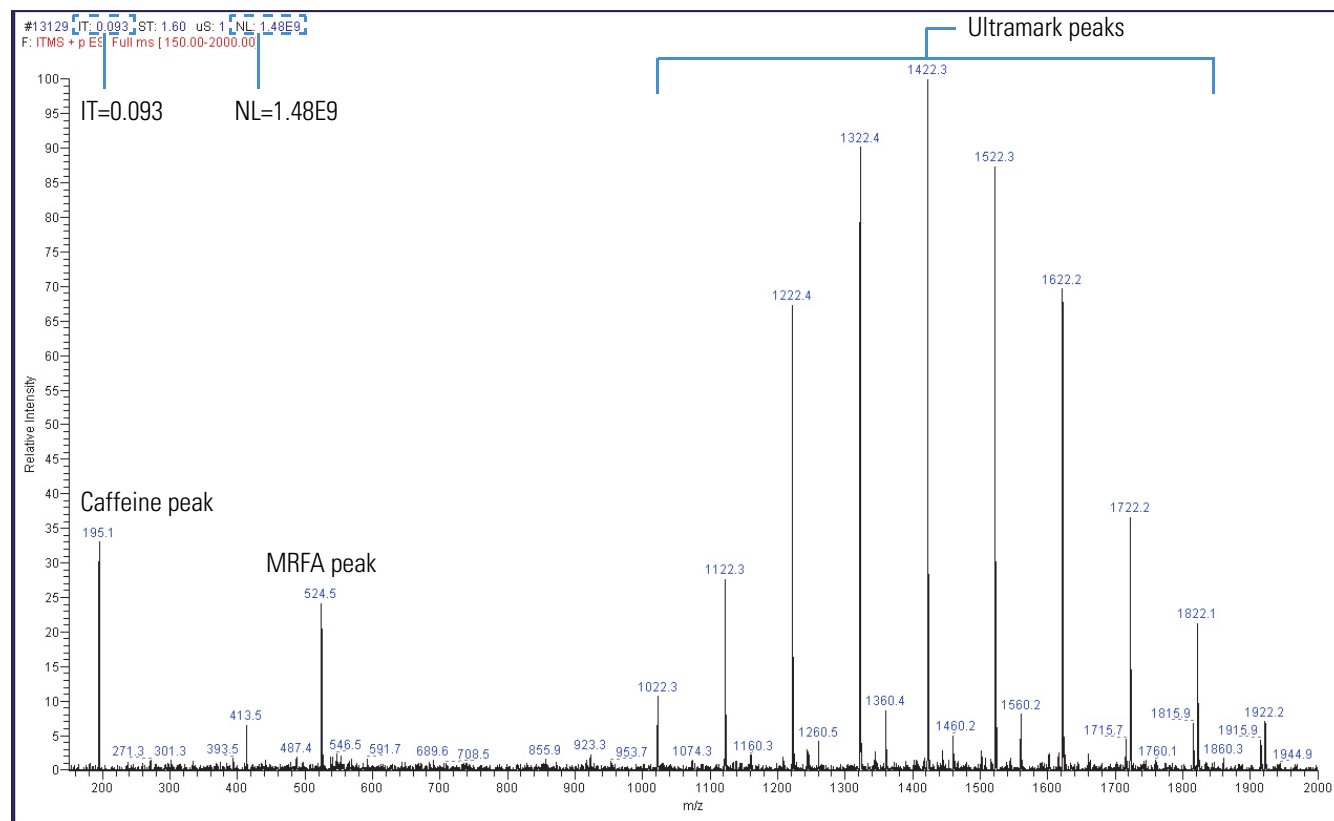


Figure 44. Spectrum of the calibration solution for the LTQ XL MS

4. At the top of the Spectrum view, check the values for the ionization time (IT) and normalization level (NL) (Figure 44).
5. Choose **Setup > ESI Source**.
6. Check the Spray Current readback.
7. Observe the values for NL and IT in the Spectrum view.

As calibration solution infuses, and the readback values fluctuate, consider the following questions about the ion current signal:

- Is the signal present?
- Is the signal stable, varying by less than about 15% from scan to scan?

To assist in assessing the signal stability, a diagnostic tool is available. In Tune Plus, choose **Diagnostics > Diagnostics**. In the Tools list, select **System Evaluation**, and then click the **API Stability Evaluation** check box.

If you answered “yes” to these questions, then the instrument is operating properly.

If you answered “no” to either of these questions, go to [step 8](#).

8. Try the following troubleshooting measures:

- Make sure that the fused-silica sample tube on an ESI probe does not extend beyond the tip of the ESI needle. For instructions about inserting the fused-silica sample tube, refer to the *LTQ Series Hardware Manual*.
- Make sure that the entrance to the ion transfer tube is clean and not blocked.
- Make sure that the solution entering the probe is free of air bubbles and that the tubing and connectors are free of leaks.
- Make sure that the flow rate of the sheath gas is 5–8 units.

If you have demonstrated that the instrument is operating properly in ESI mode, you are now ready to tune and calibrate the instrument. Leave the instrument as it is, and go to the next topic, “[Tuning the Mass Spectrometer Automatically in ESI Mode.](#)”

Tuning the Mass Spectrometer Automatically in ESI Mode

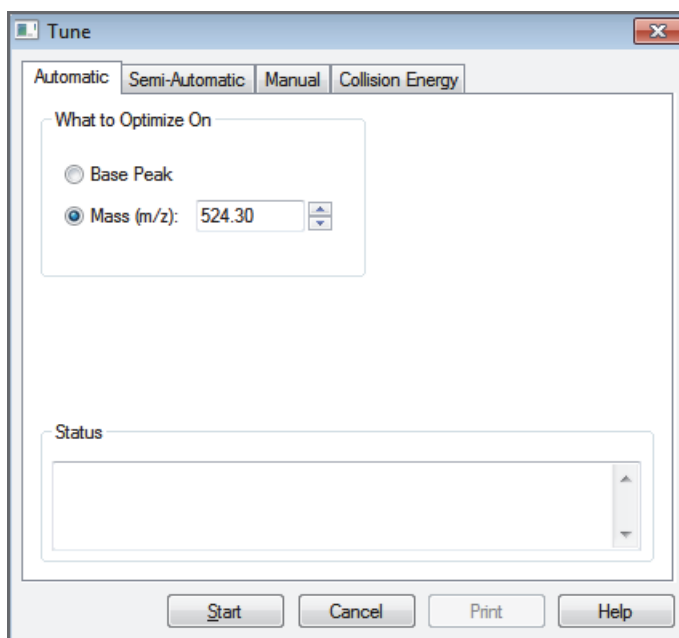
To optimize important parameters, tune the LTQ Series MS automatically in ESI mode.

❖ To automatically tune the instrument in ESI mode



1. In the Tune Plus window, click the **Tune** button to open the Tune dialog box ([Figure 45](#)).

Figure 45. Tune dialog box showing the Automatic page



2. Under What to Optimize On, select the **Mass (m/z)** option.
3. To optimize the transmission of ions for a specific peak in the mass spectrum of the ESI calibration solution, enter, for example, **524.3** in the Mass (m/z) box.

The instrument optimizes the tune on the peak at m/z 524.3. However, you can optimize the tune on any peak in the mass spectrum of the ESI calibration solution.

4. Click **Start**.

A message appears:

Please ensure that the 500 microliter syringe is full.

5. Make sure that the syringe contains at least 450 μL of calibration solution.

The Chemical Kit contains the ESI calibration solution

6. Click **OK**.



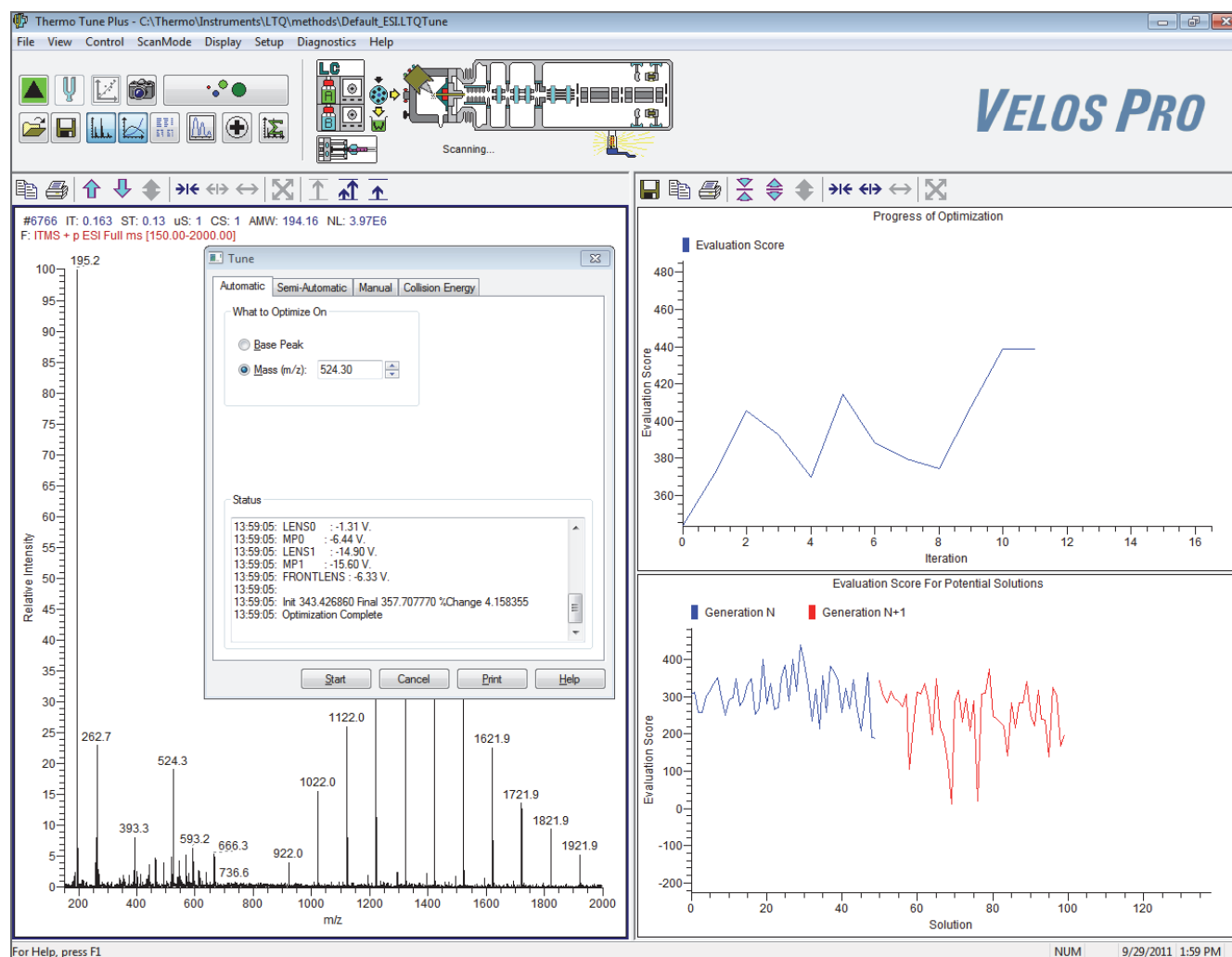
7. In the Tune Plus window, click the **Display Graph View** button.
8. Observe the Tune Plus window and the Tune dialog box.

While automatic tuning is in progress, the instrument displays the results of various adjustments in the Spectrum and Graph views in the Tune Plus window and displays various messages under Status in the Tune dialog box. [Figure 46](#) shows the Velos Pro spectrum.

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Tuning the Mass Spectrometer Automatically in ESI Mode

Figure 46. Tune Plus window and Tune dialog box to set the Velos Pro MS on m/z 524.3



When the Optimization Complete message appears in the Status box of the Tune dialog box, it should show a positive increase in the signal.

9. After the automatic tuning process ends, choose **Setup > ESI Source**.
10. Observe the post-tune settings for the ESI source.
11. Choose **Setup > Ion Optics**.
12. Observe the post-tune settings for the ion optics.

This completes the tuning in ESI mode by using the calibration solution.

Saving the ESI Tune Method

You can save the parameters that you just set in a tune method specific to the particular analyte and solvent flow rate. Recall the tune method and use it as a starting point for optimizing the instrument on a different analyte or at a different flow rate.

Note You must save the tune method while the instrument is on.

❖ To save the ESI tune method



1. In the Tune Plus window, click the **Save** button.
2. Browse to the **C:\Thermo\Instruments\LTQ\methods** folder.
3. In the File Name box, type a name, such as **ESImyTune**, to identify the tune method.
4. Click **Save**.

Calibrating Automatically in the Normal Mass Range

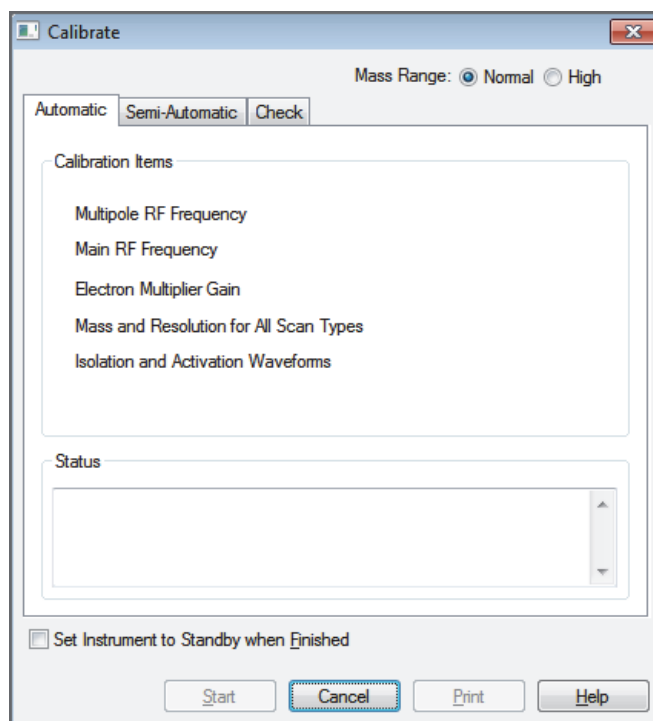
Note To calibrate the instrument in the high mass range, see [Appendix C, “High Mass Range Calibration.”](#)

❖ **To automatically calibrate the instrument in the normal mass range**



1. In the Tune Plus window, click the **Calibrate** button to open the Calibrate dialog box ([Figure 47](#)).

Figure 47. Calibrate dialog box showing the Automatic page



2. Specify the instrument state when the calibration is completed:
 - To have the instrument switch to Standby mode after the calibration, click the **Set Instrument to Standby when Finished** check box.
 - To have the instrument remain in the On mode after the calibration, clear the **Set Instrument to Standby when Finished** check box.
3. Click **Start**.

A message appears:

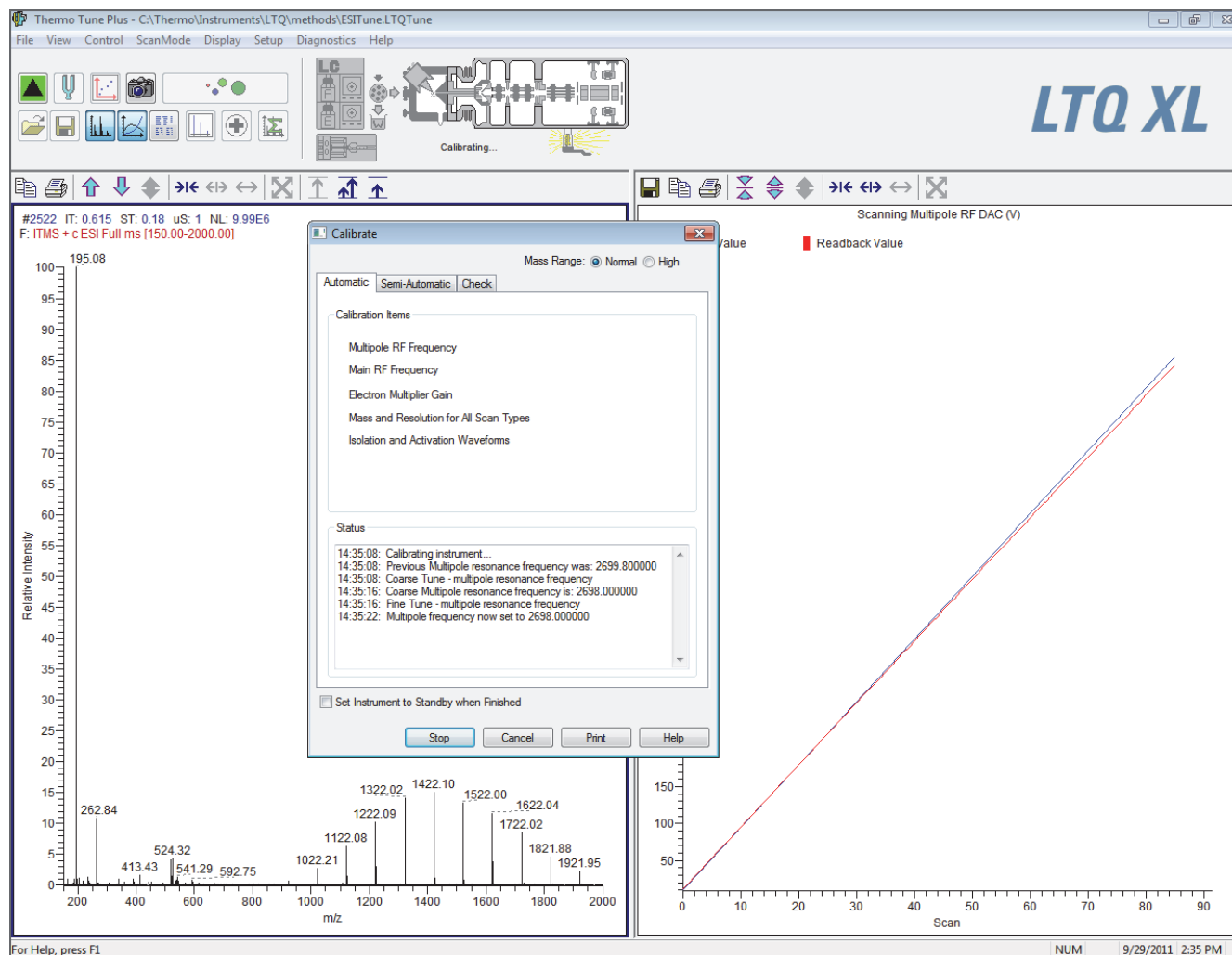
Please ensure that the 500 microliter syringe is full.

4. Make sure that the syringe contains at least 450 μL of the calibration solution.
The Chemical Kit contains the ESI calibration solution.

5. Click **OK**.
6. Observe the Tune Plus window and the Calibrate dialog box.

While the automatic calibration is in progress, test results appear in the Spectrum and Graph views and messages appear in the Status box of the Calibrate dialog box (Figure 48).

Figure 48. Tune Plus window and Calibrate dialog box for the LTQ XL MS



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Calibrating Automatically in the Normal Mass Range

The automatic calibration procedure typically takes 20–30 minutes.

After the calibration procedure ends, the full-scan ESI mass spectrum appears in the Spectrum view (Figure 48 on page 79).

- If a calibration item is successful, the instrument saves the new calibration values automatically to the master calibration file on the hard disk drive.
- If a calibration item fails, try calibrating on that item again after ensuring that the spray is stable, the solution flow rate is sufficient, and all the ions in the calibration solution are present with adequate signal-to-noise ratios. Use the semi-automatic calibration page to calibrate the specific item that failed.

Tip After the automatic calibration, green checks , red X marks , or both appear in the Result column of the Semi-Automatic page of the Calibrate dialog box. A green check indicates a successful calibration and a red X indicates a failed calibration for the corresponding item. The Status box at the bottom of the Calibrate dialog box provides additional information.

- If repeated failures occur, consider clearing the ZoomScan Mode option.

After all the calibration items are successful, the instrument is properly tuned and calibrated for low-flow experiments. A successful calibration exhibits adequate intensities of the following calibrant ions at the correct masses:

- For the Velos Pro MS: m/z 74, 138, 195, 524, 1222, 1522, and 1822
- For the LTQ XL and LXQ MSs: m/z 195, 524, 1222, 1522, and 1822

In many cases, fine tuning on the particular analyte is not necessary if the intensity of these ions is sufficient.

Note To calibrate the instrument in the high mass range, see Appendix C, “High Mass Range Calibration.”

Before tuning with an analyte, you must clean the instrument as described in Chapter 8, “Cleaning the Mass Spectrometer After Tuning and Calibrating.”

Tip If you plan to run analytical samples in high-flow ESI mode (by using flow rates of 50–1000 $\mu\text{L}/\text{min}$), optimize the tune further by following the procedures in Chapter 4, “Tuning with an Analyte in ESI Mode,” on page 81.

Tuning with an Analyte in ESI Mode

This chapter describes how to tune the LTQ Series MS in ESI mode by using an analyte. An automated procedure optimizes the sensitivity of the mass spectrometer to the specific analyte. You can often use the customized tune methods available for each instrument type without further tuning of the instrument, as they are optimized for a wide range of applications.

Contents

- [Setting Up the Inlet for High-Flow Infusion in ESI Mode](#)
- [Setting Up the Mass Spectrometer to Tune with an Analyte in ESI Mode](#)
- [Tuning the Mass Spectrometer Automatically](#)
- [Saving the ESI Tune Method](#)

The following instrument parameters significantly impact signal quality in ESI mode:

- Ion transfer tube voltage (LTQ XL and LXQ MSs only)
- Electrospray voltage
- Tube lens voltage (LTQ XL and LXQ MSs only)
- Heated ion transfer tube temperature (voltage)
- Gas flow rates for the API gases
- S-lens rf level (Velos Pro MS only)

The settings for these parameters depend on the solvent flow rate and target analyte composition. In general, fine-tune the instrument whenever you change the solvent flow rate conditions of the particular application. In this procedure, you use a low-flow tune method (or the tune file that you created for the analyte by using direct infusion) as a starting point, and then further optimize the parameters by using an automated procedure. The automatic procedure adjusts the voltages applied to the ion optics until the ion transmission of the analyte is maximized.

Note Based on the LC flow rate of the experiment, specify the value of each of the following tuning parameters: the instrument's sheath, auxiliary, and sweep gas pressures; ESI needle (or "spray") voltage; ion transfer tube temperature; and probe position. Automatic tuning sets the values of the parameters not mentioned here.

The ion transfer tube is heated to maximize the ion transmission to the instrument. For ESI only, set the ion transfer tube temperature proportional to the flow rate of the solution. Refer to the Tune Plus Help for guidelines on the operating parameters.

Note If performing the experiment at a flow rate less than 10 $\mu\text{L}/\text{min}$, and you can obtain the results without optimizing the instrument on the particular analyte, see [Chapter 5, "Acquiring ESI Sample Data by Using Tune Plus,"](#) for instructions about acquiring sample data.

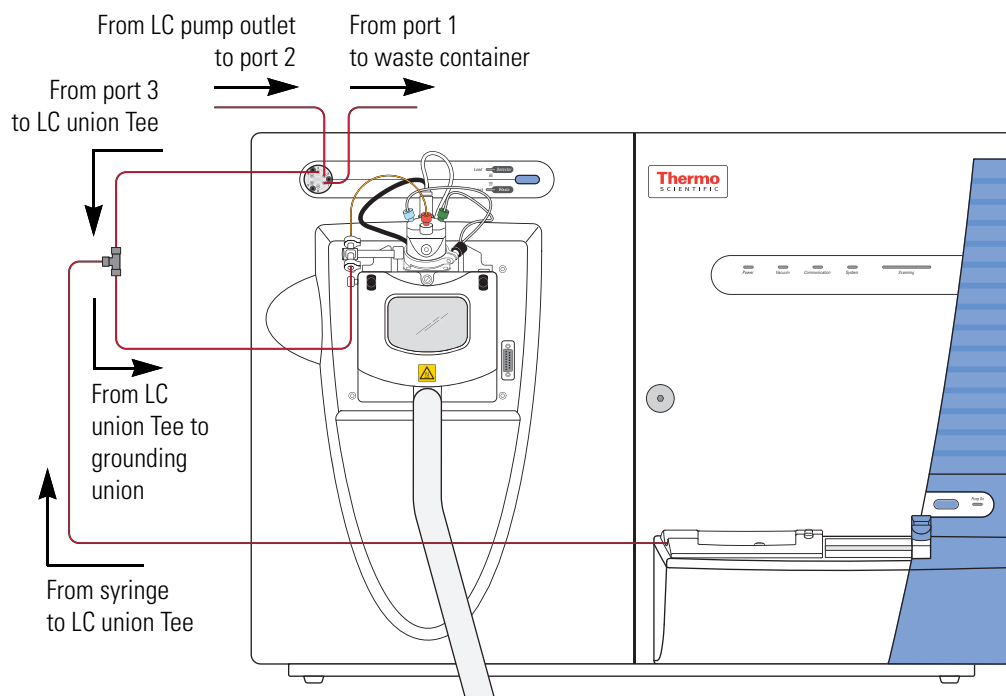
Before optimizing the tune for the analyte, make sure that the instrument has been calibrated within the last 3 months. If the system needs to be calibrated, follow the procedures in [Chapter 3, "Automatic Tuning and Calibration in ESI Mode."](#)

IMPORTANT If you make changes to the tune parameters and do not save them before switching to another API source probe, the changes are lost. In addition, when you switch from one API source probe to another, the Tune Plus application automatically loads the last saved tune parameters for the newly installed API source probe and overwrites the tune parameters from the previously installed API source probe.

Setting Up the Inlet for High-Flow Infusion in ESI Mode

This section describes how to set up the LTQ Series MS to infuse a sample solution with the syringe pump into the solvent flow from an LC pump. [Figure 49](#) shows the plumbing connections to introduce sample by high-flow infusion.

Figure 49. Plumbing setup for LC/ESI/MS infusion into the solvent flow from an LC pump



To connect the inlet plumbing for high-flow infusion analyses, use the parts listed in [Table 5](#), which are in the instrument accessory kit.

Table 5. Inlet plumbing parts for high-flow infusion analysis (Sheet 1 of 2)

Description	Part number
Grounding union (ESI mode)	00101-18182
LC union Tee	00101-18204
LC union	00101-18202
PEEK fingertight fittings and ferrules or stainless steel fittings and ferrules for conical 10-32 receiving ports and 1/16 in. OD tubing:	
• PEEK ferrule	00101-18196
• PEEK fitting	00101-18081
• Stainless steel nut	2522-0066
• Stainless steel ferrule	2522-3830

Table 5. Inlet plumbing parts for high-flow infusion analysis (Sheet 2 of 2)

Description	Part number
Red PEEK tubing (0.005 in. ID, 1/16 in. OD)	00301-22912
Teflon tubing or Injection Port filler	00301-22915
	00101-18206

To make the plumbing connections for ESI sample introduction from the syringe pump into solvent flow from an LC pump, follow these procedures:

- [Setting Up the Syringe Pump](#)
- [Connecting the Plumbing to Introduce Sample by High-Flow Infusion](#)

Setting Up the Syringe Pump

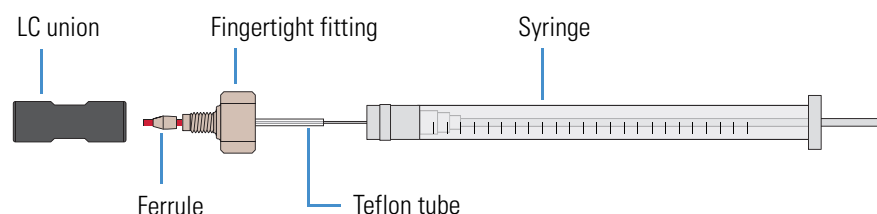
❖ To set up the syringe pump



CAUTION Sharp object. The syringe needle can puncture your skin. Handle it with care.

1. Load a clean, 500 μ L Unimetrics syringe with the 100 pg/ μ L reserpine stock tuning solution or the analyte.
2. Hold the plunger of the syringe in place and carefully insert the tip of the syringe needle into the end of a 4 cm (1.5 in.) Teflon tube with a fingertight fitting and a ferrule to the (black) LC union ([Figure 50](#)).

Figure 50. Plumbing connections for the syringe



3. Place the syringe into the syringe holder of the syringe pump.
4. Use a fingertight fitting and ferrule to connect an infusion line of red PEEK tubing between the LC union and the grounding union held by the grounding bar of the API source ([Figure 39](#) on [page 66](#)).
5. While squeezing the blue release buttons on the syringe pump handle, push the handle forward until it just contacts the syringe plunger.

Connecting the Plumbing to Introduce Sample by High-Flow Infusion

Follow these procedures to connect solvent flow from the syringe pump and the LC pump to the API source:

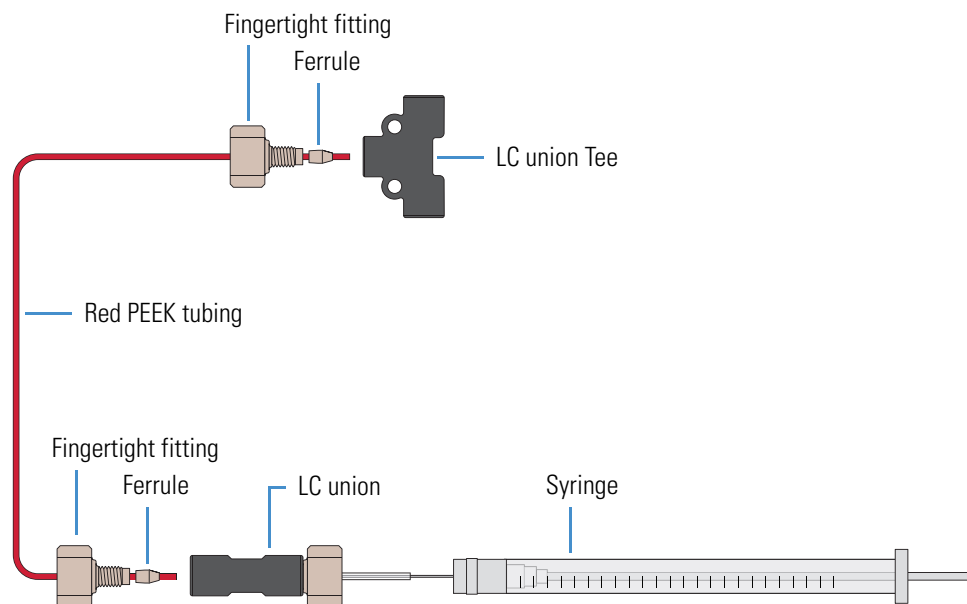
1. To connect the syringe to the LC union Tee
2. To connect the LC union Tee to the divert/Inject valve
3. To connect the LC pump to the divert/inject valve
4. To connect the divert/ inject valve to a waste container
5. To connect the LC union Tee to the grounding union
6. To connect the grounding union to the ESI probe sample inlet

❖ To connect the syringe to the LC union Tee

1. Set up the syringe pump (see [page 84](#)).
2. Use a fingertight fitting and a ferrule to connect the red PEEK infusion line to the free end of the LC union that connects to the syringe.
3. Use a fingertight fitting and a ferrule to connect the other end of the red PEEK infusion line to the LC union Tee.

[Figure 51](#) shows the fittings required to connect the LC union to the union Tee.

Figure 51. Plumbing connections between the LC union Tee and the LC union



❖ **To connect the LC union Tee to the divert/Inject valve**

1. Use a fingertight fitting and a ferrule to connect a length of red PEEK tubing to port 3 of the divert/inject valve. Or, use a stainless steel nut and ferrule to connect the tubing.
2. Use a fingertight fitting and a ferrule to connect the other end of the tubing to the free end of the LC union Tee (Figure 52 and Figure 53).

Figure 52. Six-port divert/inject valve connections

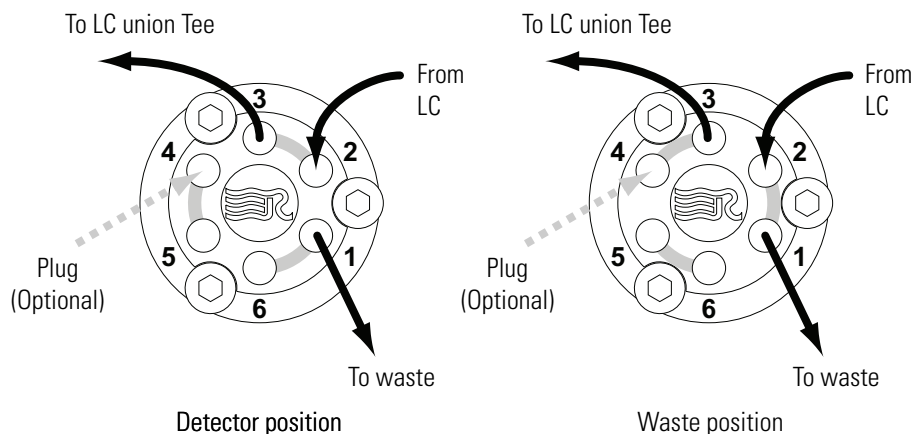
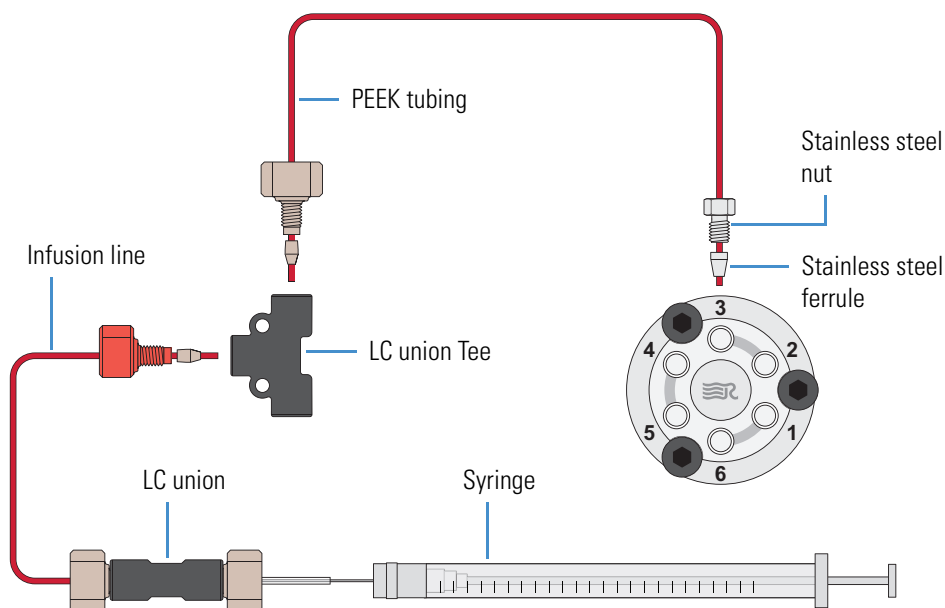


Figure 53. Connections between the LC union Tee and the divert/inject valve



❖ **To connect the LC pump to the divert/inject valve**

1. Use a fingertight fitting and a ferrule to connect a length of PEEK tubing to port 2 of the divert/inject valve (Figure 52 on page 86).
2. Use an appropriate fitting and ferrule to connect the other end of the tubing to the outlet of the LC.

❖ **To connect the divert/ inject valve to a waste container**

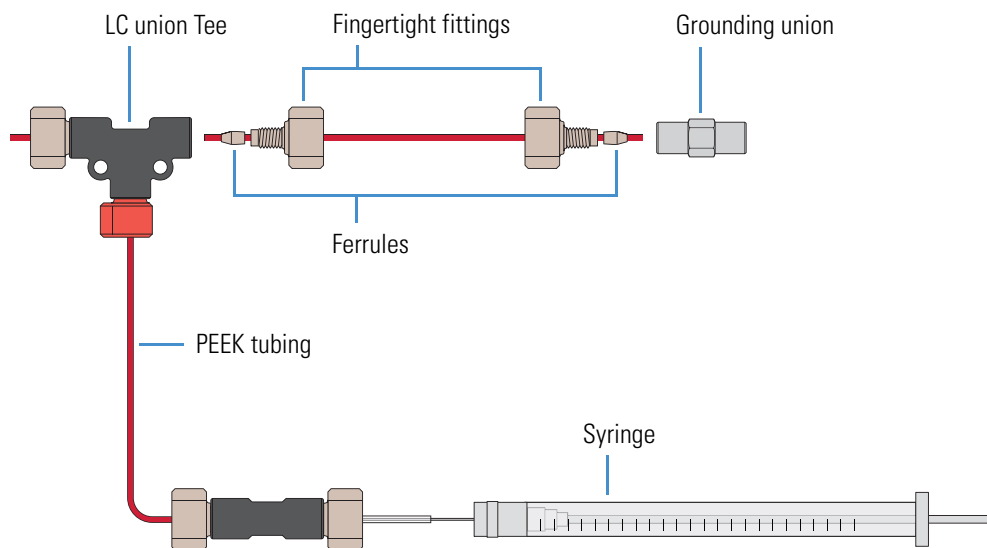
1. Use a fingertight fitting and a ferrule to connect a length of red PEEK tubing to port 1 of the divert/inject valve (Figure 52 on page 86).
2. Insert the other end of the tubing into a suitable waste container.

❖ **To connect the LC union Tee to the grounding union**

1. Use a fingertight fitting and a ferrule to connect one end of a length of red PEEK tubing to the LC union Tee (Figure 55).

The grounding union slides into the grounding bar on the API source as shown in Figure 54. For instructions about connecting the grounding union to the ESI probe sample inlet, refer to the *LTQ Series Hardware Manual*. Figure 56 shows the complete connection.

Figure 54. Connections between the LC union Tee and the grounding union (ESI probe)

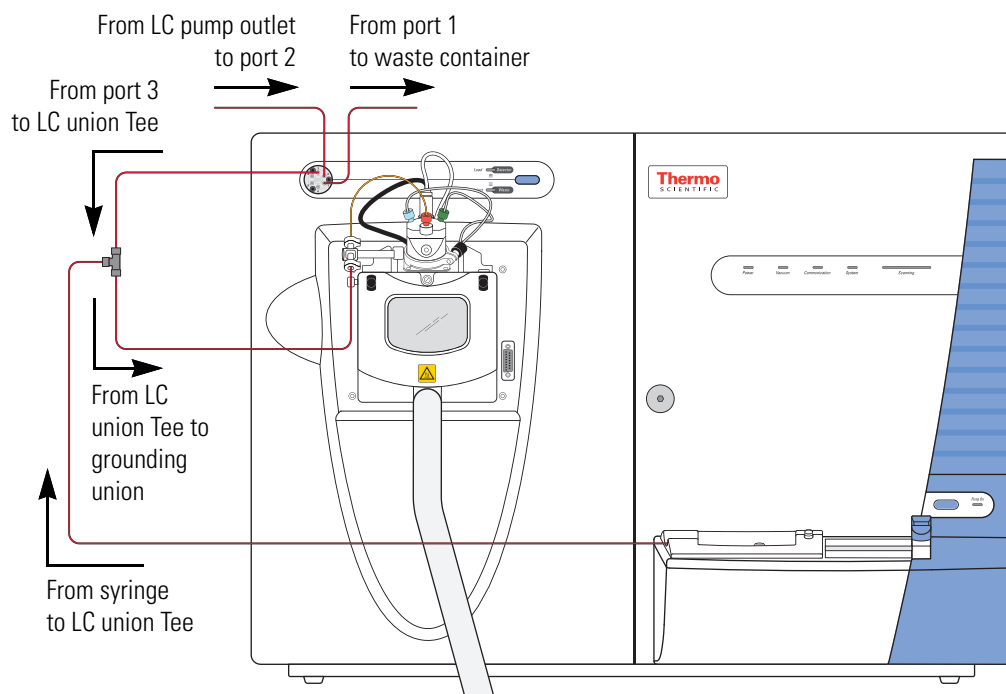


2. Use a fingertight fitting and a ferrule to connect the other end of the tubing to the grounding union that is held by the grounding bar of the API source (Figure 54 on page 87).

4 Tuning with an Analyte in ESI Mode

Setting Up the Inlet for High-Flow Infusion in ESI Mode

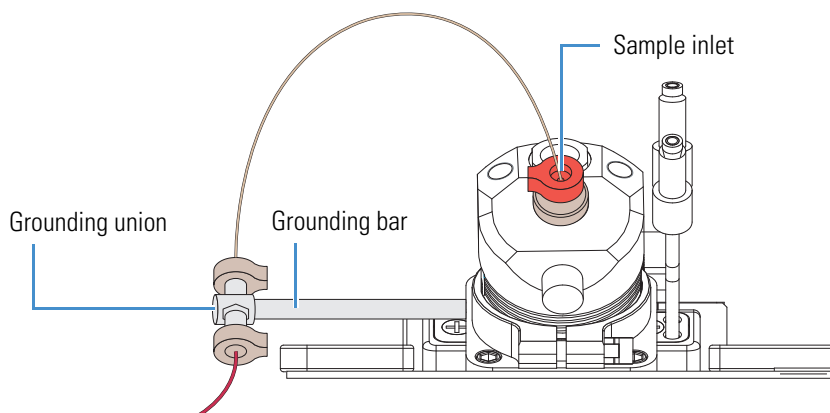
Figure 55. Plumbing setup for ESI/MS sample introduction with high-flow infusion



❖ To connect the grounding union to the ESI probe sample inlet

For instructions about connecting the PEEK safety sleeve and fused-silica sample tube from the grounding union to the ESI probe sample inlet (Figure 56), refer to the *LTQ Series Hardware Manual*.

Figure 56. Grounding union connected to the sample inlet of the ESI probe



Setting Up the Mass Spectrometer to Tune with an Analyte in ESI Mode

You can tune the LTQ Series MS in ESI mode by using a high solvent flow of the 100 pg/μL reserpine tuning solution or a solution of the analyte. Follow these procedures:

1. To open a stored tune method
2. To define the scan parameters, scan type, and scan polarity for the analyte
3. To set the data type and ion polarity mode

❖ To open a stored tune method

1. Open the Tune Plus window (see [page 32](#)).
2. Click the **On/Standby** button to select the **On** mode.



The instrument applies high voltage to the ESI probe and begins scanning. A real-time display begins in the Spectrum view.

3. Open **ESImyTune**, or the file name used for your saved tune method from Chapter 3, as follows:



- a. Click the **Open** button.
- b. Browse to the **C:\Thermo\Instruments\LTQ\methods** folder, and then select the tune method you saved.
- c. Click **Open**.

Tune Plus downloads the tune method parameters to the instrument, and the title bar in the Tune Plus window lists the name of the current tune method, for example, *C:\Thermo\Instruments\LTQ\ESImyTune*.

4 Tuning with an Analyte in ESI Mode

Setting Up the Mass Spectrometer to Tune with an Analyte in ESI Mode

❖ To define the scan parameters, scan type, and scan polarity for the analyte



1. In the Tune Plus window, click the **Define Scan** button to open the Define Scan dialog box.

Figure 57 shows the typical settings for acquiring reserpine data by using the selected ion monitoring (SIM) scan type.

Figure 57. Define Scan dialog box for acquiring reserpine SIM data in ESI mode

n	Parent Mass (m/z)	Act. Type	Iso. Width (m/z)	Normalized Collision Energy	Act. Q	Act. Time (ms)
2		CID	1.0	0.0	0.250	10.00

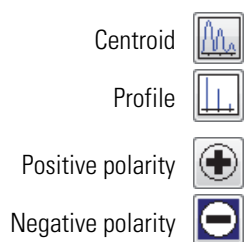
#	Center Mass (m/z)	Width (m/z)
1	609.20	2.00
2		

2. Under Scan Description, do the following:
 - In the Mass Range list, select **Normal**.
 - In the Scan Rate list, select **Normal**.
 - In the Scan Type list, select **SIM**.
3. Under Scan Time, do the following:
 - In the Microscans box, enter **2**.
 - In the Max. Inject Time (ms) box, enter **200.000**.
4. Under Source Fragmentation, clear the **On** check box.

Note Selecting the Source Fragmentation check box allows collision-induced fragmentation to occur in the API source. Because collision-induced fragmentation performed in the API source is not very specific, this feature is rarely used.

5. Under Scan Ranges, do the following:
 - a. In the Input list, select **Center/Width**.
 - b. In the Center Mass (m/z) column, type the mass of the analyte to set the center mass for the scan range. If tuning on reserpine, type **609.20** for the center mass.
 - c. In the Width (m/z) column, type **2.00**.
6. Verify the scan settings:
 - If tuning on reserpine, make sure that the settings in the Define Scan dialog box match those shown in [Figure 57](#) on [page 90](#).
 - If using another analyte, make sure that the settings in the Define Scan dialog box are appropriate for the analyte.
7. Click **OK**.

❖ **To set the data type and ion polarity mode**



1. In the Tune Plus window, click the **Centroid/Profile** button to select the **profile** data type.
2. Click the **Positive/Negative** button to select the **positive** ion polarity mode.

Note In the positive ion polarity mode, the instrument scans for positive ions. For basic compounds, such as amines, low pH solutions favor the formation of positive ions. Reserpine can form a positive ion $[(M + H)^+ m/z = 609]$ in low pH solutions.

This completes the instrument setup with an analyte in ESI mode.

Tuning the Mass Spectrometer Automatically

To maximize the transmission of ions for the analyte at a specified solvent flow rate, run the automatic tuning procedure and optimize on the target mass-to-charge ratio.

Before optimizing the tune of the LTQ Series MS on a particular mass-to-charge ratio and creating a tune method for the analyte, for best results, first calibrate the mass accuracy of the instrument. The automated calibration procedure takes approximately 20 minutes. Or, for best results in a limited amount of time, infuse the ESI calibration solution (or an analyte that you have mass spectral data for), and visually check the mass spectrum for mass accuracy. For information about calibrating the mass accuracy, see [“Calibrating Automatically in the Normal Mass Range”](#) on [page 78](#).

Note The following are the most important parameters that affect signal quality during ESI operation:

- (For the Velos Pro MS) Electrospray voltage, ion transfer tube temperature and voltage, S-lens rf level, gases, and solution flow rate
- (For the LTQ XL and LXQ MSs) Electrospray voltage, ion transfer tube temperature and voltage, tube lens voltage, gases, and solution flow rate

The most important parameters for long-term performance are the voltage gradients between MP00, MP0, and the front lens voltage. Automatic tuning ensures a sufficient voltage gradient so that the instrument continues to perform well under extended usage.

If any of these parameters change, you must reoptimize the instrument parameters. Use the Semi-Automatic tune procedure to tune the instrument on individual parameters.

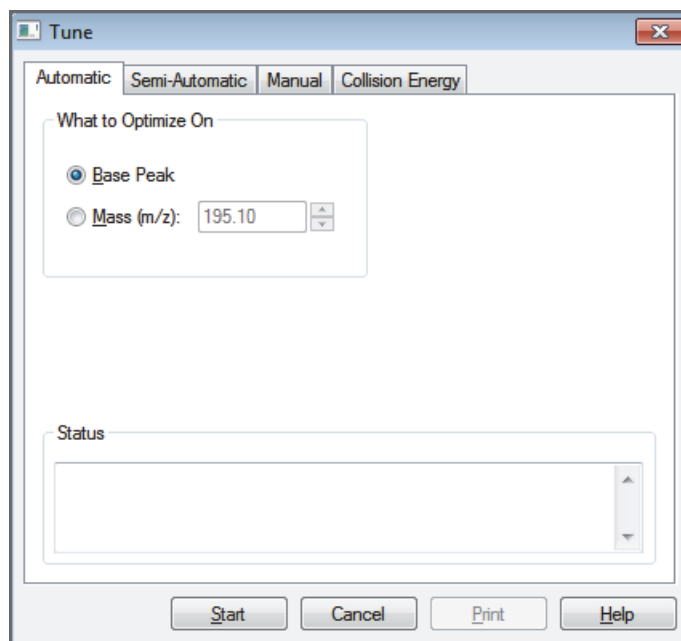
The following procedure uses a 100 pg/mL solution of reserpine to optimize the LTQ Series automatically on the reserpine peak at m/z 609.2 at a flow rate of 0.4 $\mu\text{m}/\text{min}$. You can substitute the reserpine solution with a sample solution that is suitable for analysis in ESI mode.

❖ **To automatically optimize the tune of the instrument for the analyte**



1. In the Tune Plus window, click the **Tune** button to open the Tune dialog box (Figure 58).

Figure 58. Tune dialog box showing the Automatic page



2. Under What to Optimize On, select the **Mass (m/z)** option.
3. In the Mass (m/z) box, enter **609.2** (or the appropriate mass of the analyte).



4. Click the **Divert/Inject Valve** button to open the Divert/Inject Valve dialog box (Figure 59).

Figure 59. Divert/Inject Valve dialog box



5. Select the **Load Detector** option, and then click **Close**.

Note When the divert/inject valve is in the Detector position, the solvent flow from the LC pump enters and exits the divert/inject valve through ports 2 and 3, respectively (Figure 52 on page 86). Port 3 is connected to the API source.

6. In the Tune dialog box, start the automatic tuning procedure as follows:

- a. Click **Start**.

A message appears:

Please ensure that the 500 microliter syringe is full.

- b. Make sure that the syringe pump contains at least 450 μL of the 100 $\text{pg}/\mu\text{L}$ reserpine tuning solution or the analyte.
- c. Click **OK**.



7. In the Tune Plus window, click the **Display Graph View** button.
8. Observe the Tune Plus window and the Tune dialog box.

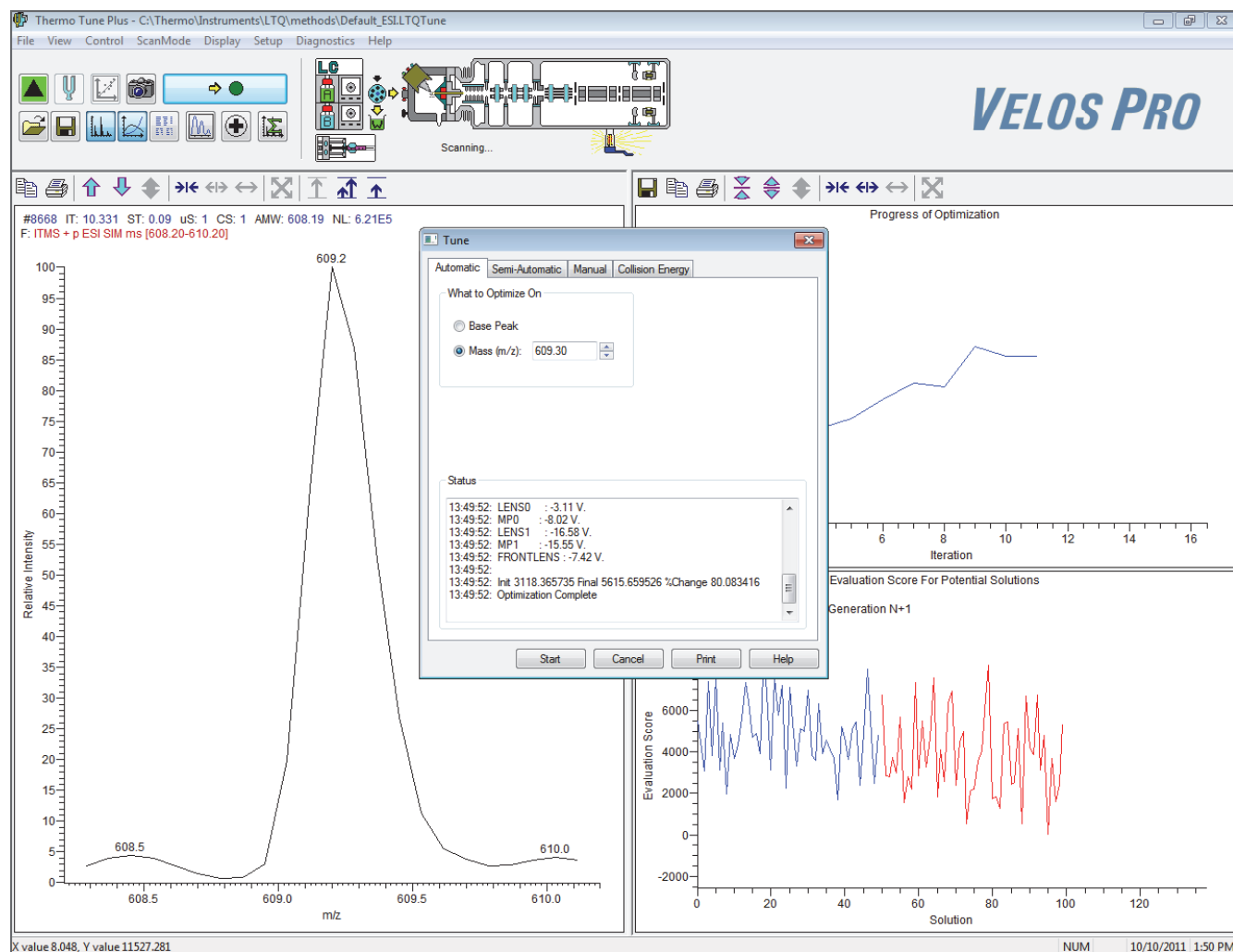
During automatic tuning, the instrument displays the results of various tests in the Spectrum and Graph views in the Tune Plus window and displays various messages under Status in the Tune dialog box.

Check that the Tune Plus window looks similar to the one shown in Figure 60.

4 Tuning with an Analyte in ESI Mode

Tuning the Mass Spectrometer Automatically

Figure 60. Tune Plus window and Tune dialog box for Velos Pro MS automatic tuning in ESI mode



This completes the tuning of the instrument in ESI mode for the compound reserpine (or the analyte).

Saving the ESI Tune Method

IMPORTANT Save the tune method before you change the type of probe.

❖ To save the ESI tune method



1. In the Tune Plus window, click the **Save** button.
2. Browse to the **C:\Thermo\Instruments\LTQ\methods** folder.
3. In the File Name box, type **reserpine** (or the name of the analyte).
4. Click **Save**.

If you tuned the instrument by using reserpine, clean it as described in [Chapter 8, “Cleaning the Mass Spectrometer After Tuning and Calibrating.”](#) If you tuned with an analyte, as described in [Chapter 5, “Acquiring ESI Sample Data by Using Tune Plus,”](#) you are ready to acquire data on the analyte.

4 Tuning with an Analyte in ESI Mode

Saving the ESI Tune Method